Longlong Xu

Multimodal LLMs; Post-Training and Alignment; Multimodal Reasoning; LLMs for AIOps

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Education

Tsinghua University, Department of Computer Science and Technology Sep. 2023 - Present

Ph.D. Student Computer Science and Technology

Research Interests: Multimodal LLMs, post-training and alignment, multimodal reasoning, LLMs for AIOps

Tsinghua University, Department of Computer Science and Technology Sep. 2018 - Jun. 2023

B.Eng. Computer Science and Technology

Representative Research

- **LLM Post-Training and Reasoning:** two ongoing first-author works on intent-aligned time-series anomaly detection and multimodal LLM reasoning over long numerical sequences; **FoundRoot**, ICSE 2026 (CCF A, co-author, link); **ChatTS**, VLDB 2025 (CCF A, 4th author, link).
- **LLM Evaluation: OpsEval**, FSE 2025 (CCF A, 3rd author, link); **Eagle**, FSE 2026 (CCF A, 4th author, link); **TechSupportEval**, IJCNN 2025 (THU B, CCF C, 4th author, link).
- **Fault Diagnosis:** one ongoing first-author work on software-change-aware fault diagnosis for online service systems; **LatentScope**, KDD 2024 (CCF A, co-author, link).
- **Time-Series Forecasting: TameR**, ICML 2026 (CCF A, first author, link); **SPRINT**, ICML 2026 (CCF A, first author, link).
- **Computer Vision:** CVPR 2023 Highlight (CCF A, 3rd author, link); IJCV 2024 (CCF A, 2nd author, link).

LLM Post-Training and Applied Reasoning Sep. 2023 - May 2026

- Proposed two ongoing first-author LLM post-training works: one targets human-intent-aligned time-series anomaly detection using data profiling, intent parsing, memory retrieval, and reinforcement learning; the other targets numerical accuracy and fine-grained localization over long sequences via active perception and post-training on multi-turn tool-use trajectories.
- Contributed to **FoundRoot** and **ChatTS**: FoundRoot builds a foundation reasoning model for fault root-cause localization with structured deep thinking; ChatTS introduces time series as a new modality for LLM-based numerical question answering and reasoning.

LLM Evaluation and Benchmark Generation Nov. 2023 - Apr. 2026

- Contributed to **OpsEval**, **Eagle**, and **TechSupportEval**, building benchmarks and automated evaluators for complex reasoning, vertical-domain question answering, operations documents, and technical support.
- Covered alerts, logs, metrics, traces, configurations, and technical documents, evaluating evidence understanding, root-cause analysis, document use, question generation, key-term matching, and step completeness.

Fault Diagnosis and Root-Cause Localization Models Apr. 2024 - Apr. 2025

- Proposed an ongoing first-author method for software-change-aware diagnosis in online services, using change-aware distribution alignment and contrastive fault clustering to maintain stable latent fault representations under concept drift.
- Researched **OTN-STFM** at Huawei, unifying optical-network performance metrics, alarms, and topology with a self-supervised spatio-temporal autoencoder; achieved 91.32%/92.98%/94.21% A@1/A@3/A@5 on 242 fault cases and was practically validated on internal backbone-network fault-localization tasks.

Robust Training and Inference-Time Scaling for Time-Series Forecasting May 2025 - May 2026

- Proposed **TameR** (link), identifying shortcut reliance on recent observations in forecasting models and improving robustness to noise, missing values, and abnormal perturbations with base-space aligned random sampling, learnable period extraction, and two-stage training.
- Proposed **SPRINT** (link), an inference-time scaling framework for long and ultra-long forecasting; without retraining or modifying the base model, it improved accuracy by 19%, reduced peak memory by 6.4×, and reduced inference time by 16.9×.

Internships

ByteDance - Algorithm Engineer May 2025 - May 2026

- Conducted research on LLM post-training, multimodal reasoning, and inference-time enhancement, producing two first-author ICML 2026 papers (**TameR**, link; **SPRINT**, link) and two first-author ongoing works.
- Built training and evaluation pipelines for multi-turn reasoning tasks, covering synthetic data generation, supervised fine-tuning, reinforcement learning / trajectory-level alignment, inference service evaluation, and multi-model baselines.
- A related intent-aligned anomaly-detection line has been validated in an internal intelligent alerting workflow.

Huawei - Algorithm Engineer Oct. 2024 - Apr. 2025

- Researched **OTN-STFM** for optical transport networks, jointly modeling performance metrics, alarms, and topology with a self-supervised spatio-temporal autoencoder for anomaly detection, fault localization, and degradation detection.
- Achieved 91.32%/92.98%/94.21% A@1/A@3/A@5 on 242 fault cases, was practically validated on internal backbone-network fault-localization tasks, and explored channel-aware clustered graph autoencoders for zero-shot localization across topologies, variable counts, and domains.

ZTE - Algorithm Engineer Nov. 2023 - Sep. 2024

- Collected and cleaned network-domain corpora, generated question-answering data, and fine-tuned general LLMs into an operations-domain LLM, improving domain QA and retrieval-augmented QA performance.
- Contributed to **OpsEval** (FSE 2025, CCF A, 3rd author) and **TechSupportEval** (IJCNN 2025, THU B, CCF C, 4th author), building evaluation data and automated QA evaluators for internal practical LLM development and assessment. (OpsEval; TechSupportEval)

eBay - Algorithm Engineer May 2023 - Oct. 2023

- Developed root-cause localization algorithms for complex internal systems by modeling causal relations among system metrics in latent space and using intervention-based identification; contributed to a KDD 2024 (CCF A) co-author paper and internal engineering validation. (link)

RealAI - Algorithm Engineer Aug. 2022 - May 2023

- Designed adversarial textured 3D meshes to bypass face-recognition defenses; published at CVPR 2023 (CCF A, 3rd author) and selected as a CVPR Highlight (top 2.5%). (link)
- Implemented robust physical face attacks with face patches by perturbing the 3D StyleGAN latent space; published in IJCV 2024 (CCF A, 2nd author). (link)

Awards

Outstanding Undergraduate Thesis, Tsinghua University	Jun. 2023
Xiaomi Scholarship	Dec. 2022
Comprehensive Excellence Scholarship, Tsinghua University	Dec. 2021, Dec. 2022
Social Work Excellence Scholarship, Tsinghua University	Dec. 2020

Skills

- Programming: Python, C/C++, Java
- Post-Training and LLM Tools: PyTorch, HuggingFace, vLLM, Open-R1, TRL, TensorFlow
- Systems and Engineering: Kubernetes, Docker, Node.js, Django, MongoDB, MySQL, Android, LaTeX